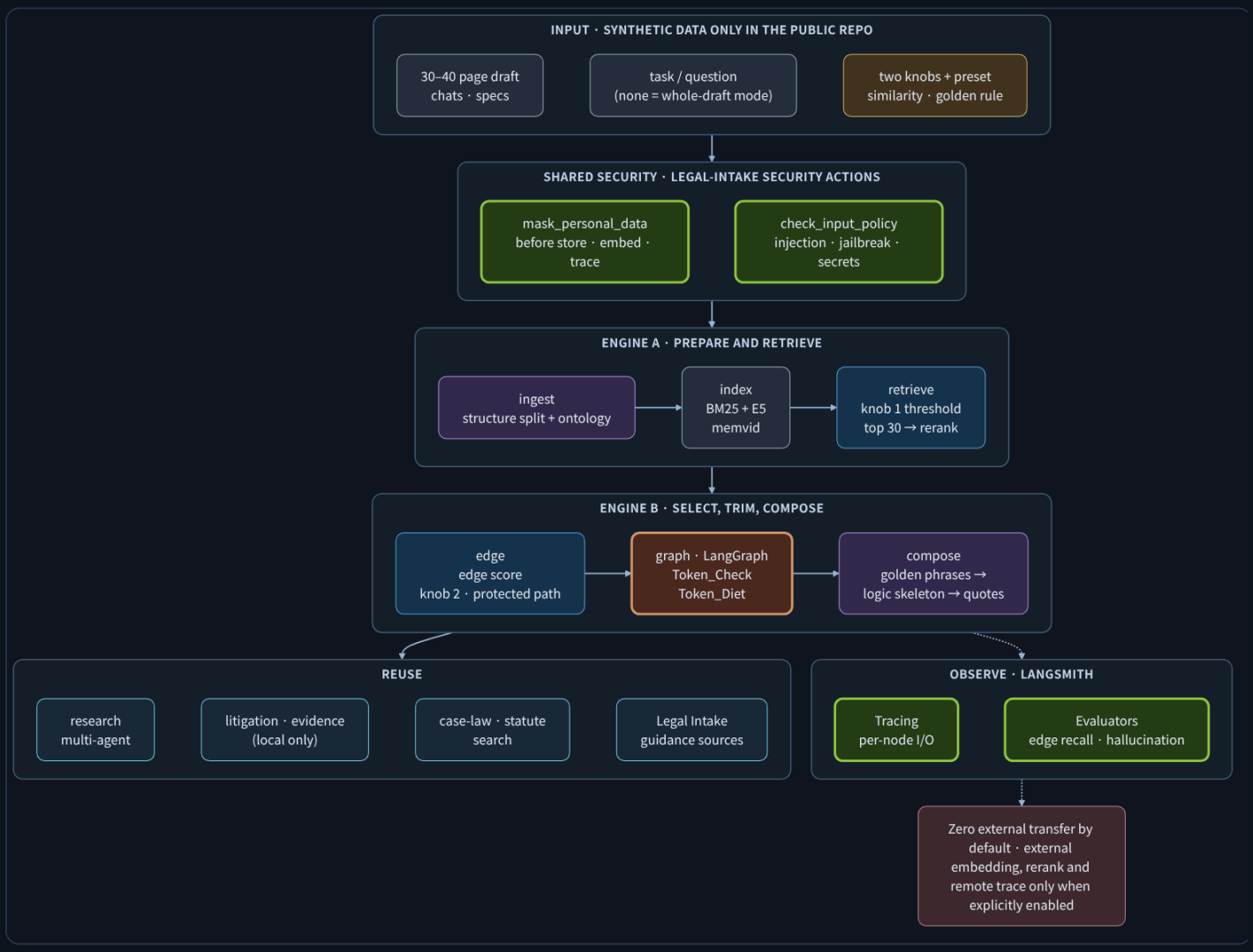
langchain-rag-research-agent

A reusable search and brief-building module: hybrid retrieval, an edge-preserving "equalizer", and a LangGraph pipeline that keeps an author's original ideas intact when a long draft is shrunk for a small local model.

한국어 README · Design review · Study guide (dark PDF)

> **Status: design v1.0 in review.** The repository is being renamed from `python-memvid-guardrail`; its former hybrid (Spring Boot + FastAPI + memvid) content is preserved in `docs/legacy/` and is moving to spring-fastapi-async-refactor. What exists today: an E5 + memvid retrieval evaluation (measured on synthetic data), a package skeleton, and a long-prompt brief prototype (fake embedder). **The Edge Equalizer itself is a design, not yet an implementation, and no real Llama or Qwen summarization quality has been measured.** Every number below is labelled *measured*, *prototype* or *example*.

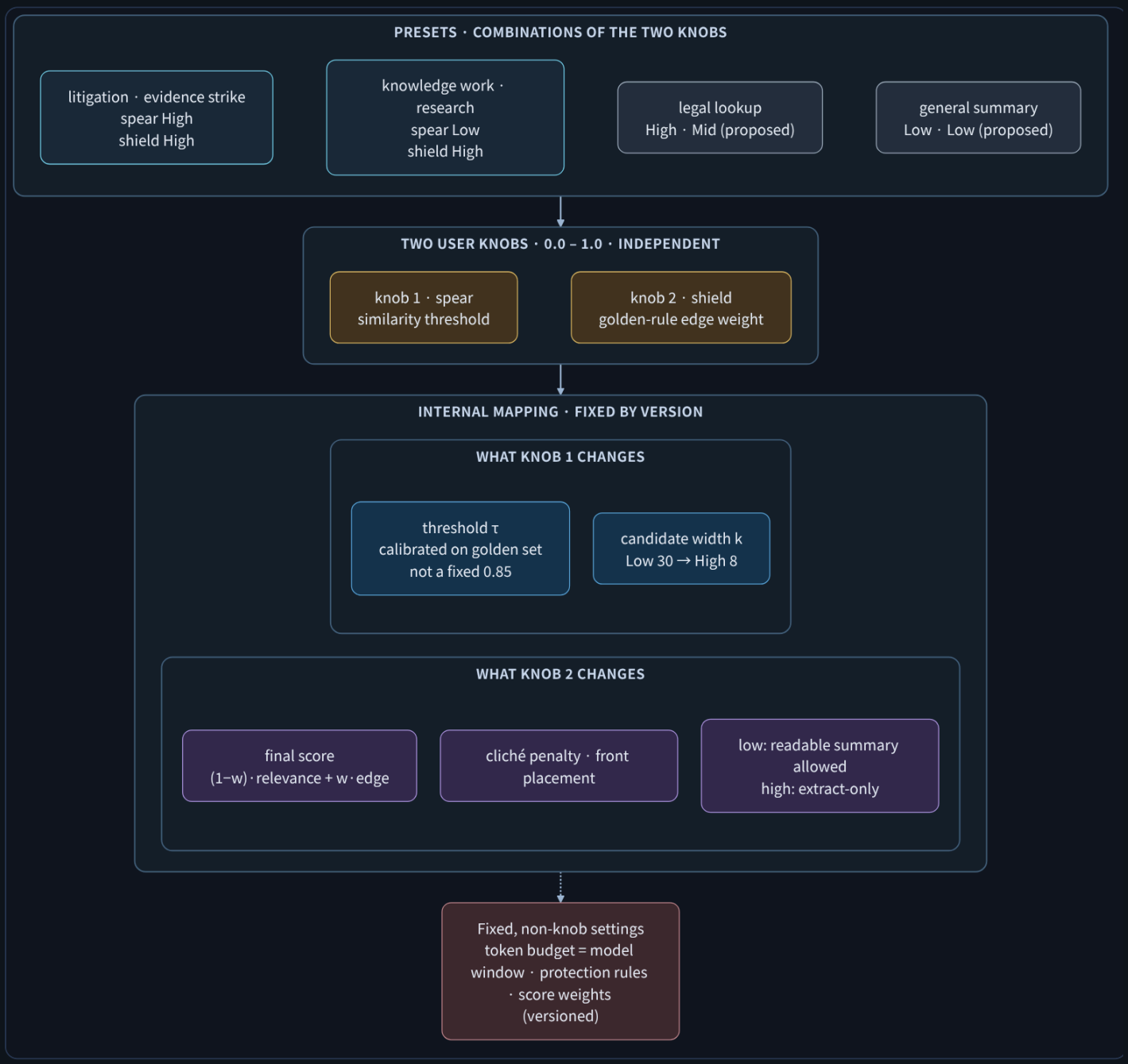


Why

A 30–40 page brainstorming draft mixes **edge sentences** (an original hypothesis, a concrete number, a cause-and-effect, a constraint) with **cliché sentences** (textbook generalities). Feeding the draft whole into a small model hits the context window, the model tends to under-use the middle of long inputs, and summaries drift toward the most common phrasing, so the edge is the first thing lost.

This module sits in front of the model. It splits the draft by structure, retrieves with keywords and meaning, scores each segment for originality, protects what must never be dropped, and trims to the model's budget, **without silently truncating**.

The Edge Equalizer (working name): two knobs



Knob	Meaning	High	Low
1. Spear: similarity threshold	how wide the candidate net is	precise, fact-focused	broad exploration
2. Shield: golden-rule edge weight `w`	how strongly original edges are kept	extract-only, clichés cut, golden phrases first	readable ontology-ordered summary

```
final_score = (1 − w) · rerank_relevance + w · edge_score
```

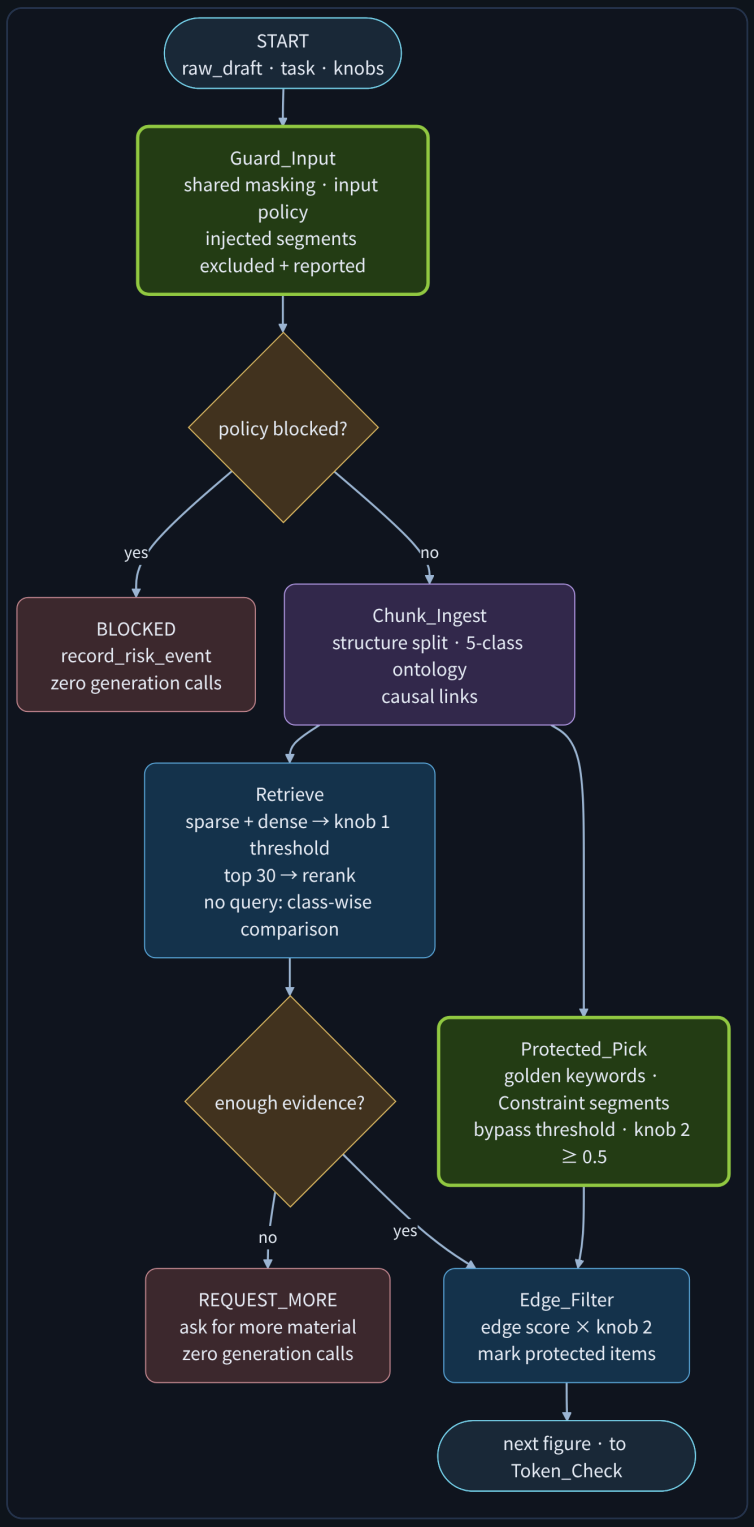
Preset	Spear	Shield	Use
Litigation / evidence strike	High	High	find exact facts, preserve subtle detail
Knowledge work / AI research	Low	High	explore widely, keep the author's own ideas
Legal lookup (proposed)	High	Mid	precise sources, key citations protected
General summary (proposed)	Low	Low	light reading

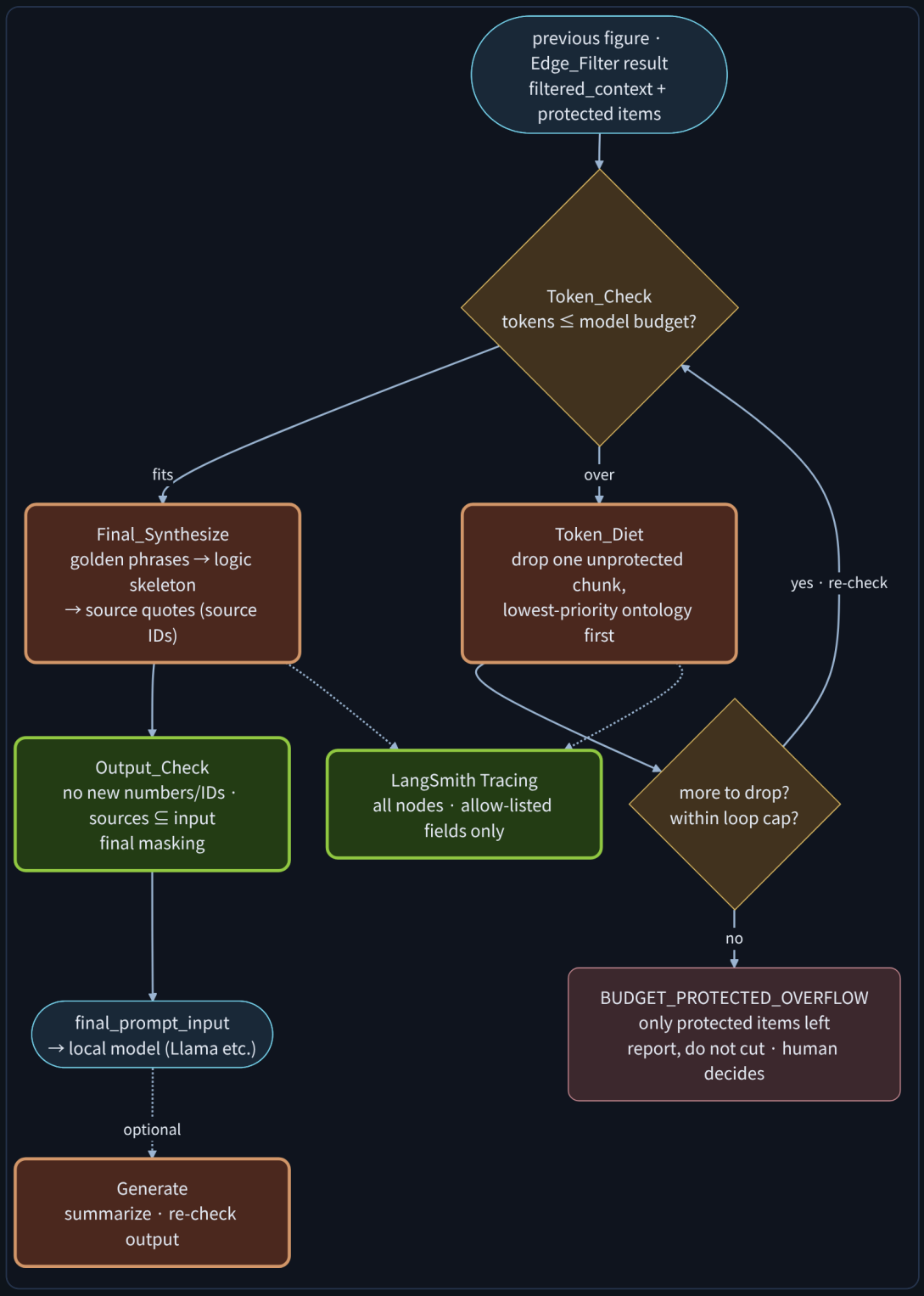
Everything that is not a knob (token budget, protection rules, score weights, diet order) is a **versioned fixed configuration**, so a result can always be explained.

Design decisions made in review (details in DESIGN_REVIEW): - The similarity threshold is **calibrated per embedding model on a golden set**, not fixed at 0.85. Measured E5 scores sit in a narrow band (lowest correct top-1 = 0.787). - A **protected-candidate path** lets golden-keyword and constraint segments bypass knob 1, so the two knobs are genuinely independent. - A reranker only measures relevance; **cliché removal is done by a separate edge score** (rarity, golden phrase, concreteness, causal markers, ontology priority, cliché penalty). - Claims are *measured*: "protected-item drop = 0" and golden-edge recall (n / total), not "zero loss".

Pipeline

LangGraph nodes: `Guard_Input → Chunk_Ingest → Retrieve / Protected_Pick → Edge_Filter → Token_Check ⇄ Token_Diet → Final_Synthesize → Output_Check`.


START
raw_draft · task · knobs
Guard_Input
shared masking · input policy
injected segments
excluded + reported
policy blocked?
yes
no
BLOCKED
record_risk_event
zero generation calls
Chunk_Ingest
structure split · 5-class ontology
causal links
Retrieve
sparse + dense → knob 1 threshold
top 30 → rerank
no query: class-wise comparison
enough evidence?
no
yes
REQUEST_MORE
ask for more material
zero generation calls
Protected_Pick
golden keywords · Constraint segments
bypass threshold · knob 2 ≥ 0.5
Edge_Filter
edge score × knob 2
mark protected items
next figure · to Token_Check



- Hybrid retrieval: BM25/TF-IDF + local multilingual E5 vectors fused by RRF, top 30, local reranker (BGE by default).
- Ontology (5 classes): `Core_Hypothesis`, `Tech_Architecture`, `Market_Positioning`, `Constraint_Guardrail`, `Action_Item` (+ internal `Unclassified`, never deleted). The parser labels with offsets and never rewrites text.
- Causal links (`causes`, `supports`, `contradicts`, `supersedes`) so a superseded decision never ranks above the current one.
- Token Diet removes unprotected segments in a fixed order. If only protected items exceed the budget it **stops with** `BUDGET_PROTECTED_OVERFLOW` instead of cutting.
- No evidence means **zero generation calls**.

Security and privacy

- **Local by default**: embeddings, reranker and generator run on the device; external providers sit behind interfaces and are opt-in per run, and forbidden in the litigation preset.
- **Masking and input policy are shared** with the security actions of legal-intake-fastapi-lab (canonical there). This repository defines the protocol and a fake implementation for tests.
- Instruction block and data block are separated; indirect-injection checks run per segment; output is rejected if it adds numbers, dates or IDs not in the source, or cites a source ID that was not supplied.
- LangSmith tracing is **off in serving code** and only used for evaluation on synthetic or masked data with an allow-listed field set.
- This public repository holds **synthetic data only**.

Open-source models (Llama and others)

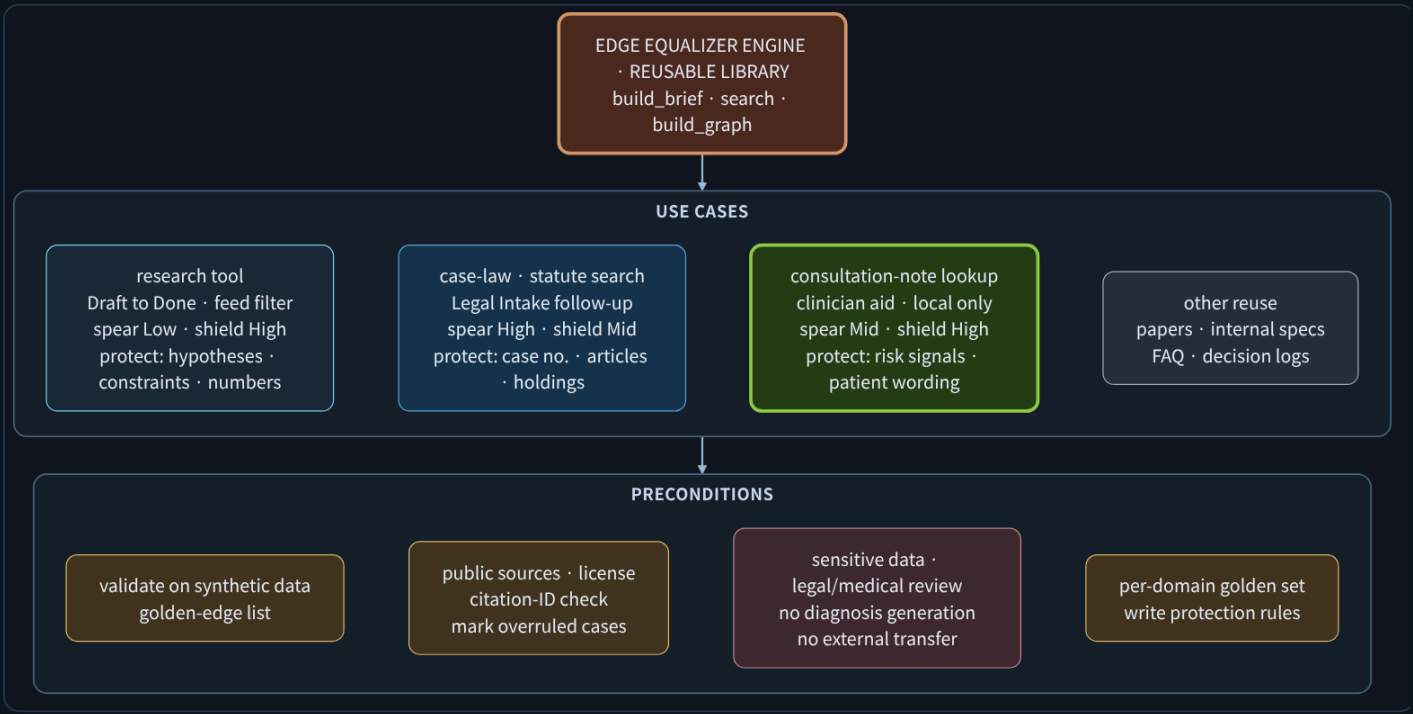
The generator, embedder and reranker are interfaces, so a local model can be swapped without touching the engine. The expected quality gain comes from better *input* (budget-fit, edge-first, extract-only, source-tagged), not from a bigger model; it is a hypothesis to be measured against five baselines (head cut, char-chunk RAG, model-direct, optional Claude-direct, this pipeline) using golden-edge recall. Qwen-family models are excluded from the default scope pending an owner decision.

What is measured, what is not

Item	Status	Basis
E5 + memvid retrieval	**measured**	synthetic 10 docs / 30 chunks: Hit@1 keyword 12/14, vector 14/14 on the compare set; 5/6 vs 6/6 on the confirm set; query embedding ≈ 0.18–0.20 s on CPU, search < 2 ms
Brief prototype (Draft PR)	**prototype**	fake embedder, 41.6 synthetic pages: recall head-cut 0.0, BM25 0.467, vector 0.445, hybrid 0.943
Ontology parser, two knobs, edge score, protected path, Token Diet	design	review complete, not implemented
Real Llama/Qwen summary quality	not run	planned in M5

Synthetic, small samples: tendencies, not proof.

Where it can be reused



Research tool (Draft to Done), public case-law and statute lookup for Legal Intake, clinician-side lookup over consultation notes, paper and spec search. The last two are the most sensitive and need legal and clinical review, local-only operation, no diagnosis or legal-judgment generation, and synthetic-only development. See the study guide.

Related repositories

Repository	Role
legal-intake-fastapi-lab	original intake design, 9 security actions, marketing hardening (≤ 11 APIs); owns masking
spring-fastapi-async-refactor	Spring Boot control plane + FastAPI + memvid hybrid (moving from this repo)
ai-multi-agent-orchestration	hub: the two multi-agent systems and the research harness

Roadmap

M0–M1 package skeleton and retrieval, reproduce the B02b result here · M2 embedding-model comparison on a golden set · M3–M4 shared security, graph and tracing, zero-egress tests · **M5 Equalizer core (ontology parser, two knobs, edge score, protected path, Token Diet) evaluated with a real local model** · M6 research-tool shadow run · M7 legal-lookup pilot on public, synthetic data.

Documents

DESIGN_REVIEW.ko.md · MODULE_DESIGN.ko.md · Study guide PDF · diagram sources · legacy hybrid README